Yuqi Zhang

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Education

- **Boston University (BU) [🏛]** **09/2026 – Present**
  Ph.D student in Astronomy, Department of Astronomy *Boston, Massachusetts, U.S.*
  - Advisor: Professor Wen Li
- **University of California, Berkeley (UCB) [🏛]** **01/2025 – 08/2025**
  Exchange student, Berkeley International Study Program (BISP) *Berkeley, California, U.S.*
  - Advisor: Doctor Orlando M. Romeo
  - GPA: 3.9/4.00
- **Southern University of Science and Technology (SUSTech) [🏛]** **08/2022 – 07/2026**
  B.S. in Geophysics, Department of Earth and Space Sciences (ESS) *Shenzhen, Guangdong, P.R.C.*
  - Advisor: Professor Shengyi Ye
  - GPA: 3.89/4.00
  - Major Courses: Mathematical Methods in Physics (100), Numerical Analysis (100), Plasma Physics (95), Continuum Mechanics (99), Fundamentals of Planetary Science (94), Electrodynamics (94), Principles of Geophysics (96), Principles of Geology (94), Principles of Applied Geophysics (95)

Publications

C=Conference, J=Journal, Prep=In Preparation, Sub=In Submission, T=Thesis

[Sub] **Y. Zhang**, et al. (2026). **Three-Dimensional Dust Distribution in the Jovian System from Juno/Waves Observations: Insights into the Halo Ring and Magnetospheric Dust**. Manuscript submitted for publication in *Astrophysics Journal*.

[Prep] W. Li, **Y. Zhang**, et al. (2026). **The Dust Environment of Mars Observed by MAVEN**.

[C] W. Li, **Y. Zhang**, et al. (2026). **The Dust Environment of Mars Observed by MAVEN**. Poster presentation in *Annual Meeting of Chinese Geoscience Union (CGU)*. 10/2026, Hangzhou, Zhejiang, P.R.C.

[C] S. Ye, W. Li, **Y. Zhang**, et al. (2026). **Long-Term Dust Impact Observations in the Martian Environment from MAVEN/LPW**. Poster presentation in *American Geophysical Union Annual Meeting (AGU)*. 12/2026, San Francisco, California, U.S.

[T] **Y. Zhang**. (2026). **On the Distribution of Dust within Jupiter System**. Bachelor's Level. 05/2026.

[C] S. Ye, W. Li, **Y. Zhang**, et al. (2025). **Electric Field Antenna Signals Induced by Dust Impacts on MAVEN Revealing Dusty Environment around Mars**. Poster presentation in *AGU*. 12/2025, New Orleans, Louisiana, U.S.

[C] **Y. Zhang**, et al. (2025). **ICME Evolution Based on Multi-spacecraft Observations**. Poster presentation in *Annual Meeting of Chinese Geoscience Union (CGU)*. 10/2025, Chengdu, Sichuan, P.R.C.

Projects

- **CME-MOSS: [Multi-Spacecraft CME/ICME Propagation and Encounter Analysis Software]** **Month/Year – Month/Year**
  Tools: Python, NASA DONKI **[]**
  - Developed an end-to-end Python framework for multi-spacecraft CME/ICME analysis, integrating DONKI catalog queries, JPL Horizons ephemerides, analytic ICME propagation, and GUI/CLI/API interfaces into one reproducible pipeline
  - Implemented Parker-spiral background-flow geometry with a two-branch drag-based propagation model (four selectable variants), capturing ICME deceleration/acceleration toward the ambient solar wind
  - Created a strict spatiotemporal wavefront-encounter detection (wavefront sweep + ICME-region membership within a tunable time window), replacing static cone-intersection tests and eliminating false-positive encounters

Honors and Awards

- Second Prize, 11th National College Students' "Innovation Cup" Geophysical Knowledge Competition [] **05/2026**
- Excellence Award, Guo Xie Birong Scholarship [] **05/2026**
- Distinguished graduate of SUSTech [] **05/2026**
- Outstanding graduate of ESS [] **04/2026**
- The Scholarship for Leadership Excellence at Shuren College (SRC) [] **12/2025**

- Third-Class Scholarship for Outstanding Students at SUSTech [🎗] **10/2025**
- First-Class Scholarship by ESS [🎗] **09/2023, 2024, 2025**
- The Bronze Prize of The 2024 University Physics Competition [🏅] **01/2025**
- First-Class Scholarship for Outstanding Students at SUSTech [🎗] **10/2023, 2024**
- The Third-Class Scholarship for Leadership Excellence at SRC [🎗] **01/2024**

School Experience

- Public Relations Committee (PR), Student Government (Student Union) of SUSTech (SUSU); PR, Sports Association (SA) of SRC **09/2022 – 10/2023**
- President, SUSU; President, SA of SRC (Restructured to SA of SUSTech in 04/2024) **10/2023 – 11/2024**
- President, Student Association of ESS **12/2024 – 12/2025**

Academic Services

- **Volunteer**: BU Astronomy Observatory Night **09/2026**